

Physical Injury Risk Prediction in Sports Using Machine Learning

A Mini Project Report

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by

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CERTIFICATE

This is to certify that the report entitled **Physical Injury Risk Prediction in Sports Using Machine Learning** submitted by **Lathika** (KTU Reg No: _____), **Sijo** (KTU Reg No: _____), **Gokul** (KTU Reg No: _____) & **Aswin** (KTU Reg No: _____), to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the mini project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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Abstract

Sports-related injuries impose significant physical, psychological, and financial burdens on athletes and organisations. This project presents an intelligent, end-to-end injury risk prediction system that fuses **wearable workload data** with **video-based biomechanical analysis** to generate a continuous *Injury Risk Score* (0–100) for each athlete before their next training session.

The data pipeline processes wearable signals — heart rate variability (HRV), sleep hours, training duration, sprint count, and intensity ratings — alongside advanced feature engineering: Acute:Chronic Workload Ratio (ACWR), 7-day rolling statistics, lag features, and athlete-specific expanding baseline normalisation. A **hybrid ensemble** combines XGBoost (60%) and a stacked LSTM (40%) with an agreement-based confidence score, achieving R^2 of 0.52 (XGBoost) and 0.41 (LSTM) on a synthetic dataset of 50 athletes across 180 days (8,850 sessions, 74 features).

A **MediaPipe full-body pose estimation** pipeline extracts 10 biomechanical signals per video frame, detects five movement phases with phase-specific risk weighting, computes a Movement Quality Score (MQS, 0–100), and classifies five injury types: ACL tear, hamstring strain, ankle instability, stress fracture, and shoulder overuse.

The complete system is deployed as a 6-page **Streamlit dashboard** with real-time alerts and a **FastAPI** REST service returning structured JSON predictions — providing scalable, interpretable, and actionable decision support for sports coaches and medical staff.

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Chapter 1

Introduction

Musculoskeletal injuries are among the most common and costly problems in professional and amateur sport. In the Premier League alone, injuries cost clubs an estimated £220 million per season in wages paid to unavailable players [1]. In India, the rise of professional leagues in football, cricket, kabaddi, and basketball has created an urgent need for scientifically grounded injury prevention tools accessible to coaching staff.

The consequences of sports injuries extend far beyond the immediate physical trauma. Athletes face prolonged rehabilitation periods, psychological distress from time away from competition, and in many cases a permanent reduction in peak performance. For professional clubs and national federations, a single serious injury to a key player can determine the outcome of an entire season or tournament. At the grass-roots level, young athletes who sustain preventable injuries during formative training years may abandon sport altogether — representing a significant loss to both individual wellbeing and national sporting development.

Despite the scale of the problem, injury prevention in most teams remains reactive rather than proactive. Physiotherapists typically intervene only after an injury has already occurred or after an athlete reports discomfort. The absence of continuous, objective monitoring means that subtle warning signals — a gradual rise in training load, declining sleep quality, asymmetric movement patterns — go undetected until they manifest as acute injuries. This gap between available data and actionable insight represents a clear opportunity for machine learning.

The proliferation of affordable wearable devices (GPS vests, smartwatches, heart

rate monitors) means that per-session physiological and load data are now routinely collected even at semi-professional levels. Simultaneously, advances in computer vision — particularly lightweight pose estimation frameworks such as MediaPipe — have made it possible to extract biomechanical information from standard video footage without laboratory-grade equipment. When these two data streams are integrated through a well-designed machine learning pipeline, they create a foundation for truly personalised, history-aware injury risk assessment.

Traditional injury risk assessment relies on subjective coach observation or expensive biomechanics laboratories with force plates and marker-based motion capture systems — infrastructure unavailable to most teams. This project is motivated by the goal of making an AI-powered injury prediction system open, accessible, and deployable on commodity hardware, closing the gap between elite sports science and everyday coaching practice.

Chapter 2

Literature Review

Early work on injury prediction used logistic regression on ACWR thresholds, establishing the “sweet spot” zone of 0.8–1.3 as low-risk [2]. Gabbett [3] popularised ACWR in team sports, though subsequent meta-analyses noted limitations in its predictive validity when used in isolation. Machine learning approaches were introduced to capture non-linear interactions between workload features that ACWR alone cannot model.

Rossi *et al.* [4] applied Random Forest and Gradient Boosting to GPS and accelerometry data from 27 professional football players, achieving an AUC of 0.76. Liu *et al.* [5] extended this with deep learning (LSTM) applied to sequential training load data, reporting improved temporal modelling. Neither study incorporated video-based biomechanical features.

On the computer vision side, Cao *et al.* [6] introduced OpenPose for real-time multi-person pose estimation. Google’s MediaPipe [7] later offered a lighter-weight, production-grade alternative supporting full-body, face, and hand landmarks at 30+ fps on mobile devices. Taborri *et al.* [8] used IMU-based knee angle measurement to predict ACL injury risk, but required laboratory-grade sensors not suitable for field use.

Commercial systems such as **Catapult Sports** and **Sparta Science** provide athlete monitoring platforms using proprietary normalisation similar to the expanding-window baseline adopted in this project — validating the approach but without open-source accessibility.

2.1 Summary of Related Works

Table 2.1: Summary of Related Works on Sports Injury Prediction

Author(s)	Method	Data Source	Performance
Hulin et al. (2016)	ACWR logistic regression	GPS wearables	AUC 0.62
Rossi et al. (2018)	Random Forest, GBM	GPS + accelerometer	AUC 0.76
Liu et al. (2020)	LSTM	Sequential load data	AUC 0.80
Taborri et al. (2021)	SVM + IMU angles	Lab IMU sensors	Acc. 84%
Proposed System	XGBoost + LSTM + MediaPipe	Wearable + video	R^2 0.52

2.2 Gaps in the Literature

- No existing open-source system integrates tabular workload features *and* video biomechanics in a unified risk score.
- Most models are trained on small proprietary datasets from professional clubs, limiting generalisability.
- Temporal modelling via LSTM is underexplored for injury prediction compared to its widespread use in other biomedical time series tasks.
- Deployed, interactive systems providing real-time coaching-facing dashboards are largely absent from published literature.

The proposed system directly addresses all four gaps.

Chapter 3

Proposed Methodology

3.1 Problem Statement

Sports injuries represent a significant challenge in competitive and recreational athletics, causing physical harm, extended absence from training, and substantial financial cost to athletes and organisations. Existing injury prevention approaches depend on manual observation, periodic physiotherapy assessments, or expensive laboratory-based biomechanics testing — none of which provide continuous, real-time, personalised risk monitoring at scale.

The core problem addressed in this project is: *Given the recent session-by-session wearable workload history of an athlete and a video clip of their training movements, how accurately can a machine learning system predict the probability of injury in the next session, and which specific injury type is most likely?*

Specifically, the system must: (a) ingest per-session wearable metrics and athlete history without data leakage between training and inference; (b) process video footage of training movements using pose estimation to extract biomechanical risk signals; and (c) fuse both streams to output a continuous Injury Risk Score (IRS) in the range $[0, 100]$, an injury type classification across five categories, and coaching recommendations — all served via a REST API and an interactive dashboard accessible to non-technical coaching staff.

Objectives

- To engineer a rich feature set from wearable workload data including ACWR, rolling statistics, lag features, and athlete-specific baseline normalisation.
- To develop a hybrid ensemble model combining XGBoost and a stacked LSTM for tabular and sequential modelling respectively.
- To build a real-time video analysis pipeline using MediaPipe Holistic for full-body pose estimation and biomechanical risk scoring.
- To compute a Movement Quality Score (MQS) that quantifies technique quality across five movement phases.
- To deploy the system as an interactive Streamlit dashboard and a FastAPI REST API.
- To compare multiple regression and classification models on a synthetic dataset of 50 athletes spanning 180 days.

Scope

The system targets field sports with high injury incidence: football, basketball, and athletics. The wearable pipeline is validated on a synthetic dataset (50 athletes, 180 days, 8,850 sessions, 74 features) generated using clinically validated injury epidemiology distributions. The video pipeline operates on standard 30 fps RGB video captured by a single fixed camera, requiring no specialised hardware beyond a smartphone or webcam.

3.2 Proposed Methodology

The proposed system integrates two complementary machine learning pipelines — a tabular workload pipeline and a video biomechanics pipeline — through a hybrid ensemble that produces a unified Injury Risk Score. The novelty of this approach lies in three aspects: (i) athlete-specific expanding-window baseline normalisation that

eliminates data leakage while capturing individual training history, (ii) the fusion of XGBoost’s feature interaction learning with an LSTM’s temporal sequence modelling through an agreement-weighted confidence mechanism, and (iii) the real-time integration of MediaPipe full-body pose estimation into a phase-aware Movement Quality Score without requiring specialised hardware.

Dataset: A synthetic dataset was constructed for 50 athletes across 180 days, yielding 8,850 training sessions. Each session record contains 74 features across the following categories:

Table 3.1: Feature Categories and Count

Feature Category	Count	Examples
Raw wearable metrics	5	HRV, sleep hours, training duration, sprint count, intensity rating
ACWR features	3	ACWR (4:1), ACWR (3:1), monotony index
7-day rolling stats	15	Rolling mean, std, min, max per metric
Lag features	21	Previous 1–3 session values per metric
Athlete baseline	10	Expanding mean/std normalised metrics
Biomechanical (MQS)	10	Knee angle variability, shoulder symmetry, torso lean, etc.
Derived indicators	10	Fatigue index, sleep debt, strain score, etc.
Total	74	

Data Pre-processing — Wearable / tabular data:

- Missing values imputed using forward-fill within each athlete’s time series.
- Athlete-specific expanding-window baseline normalisation: for session t of athlete a ,

$$x_{a,t}^{\text{norm}} = \frac{x_{a,t} - \mu_{a,1:t-1}}{\sigma_{a,1:t-1} + \epsilon} \quad (3.1)$$

where $\mu_{a,1:t-1}$ and $\sigma_{a,1:t-1}$ are the expanding mean and standard deviation up to but not including session t , preventing data leakage.

- ACWR computed as:

$$\text{ACWR} = \frac{\text{Acute Load (7-day mean)}}{\text{Chronic Load (28-day mean)}} \quad (3.2)$$

Data Pre-processing — Video / pose data:

- Input video downsampled to 15 fps for computational efficiency.

- MediaPipe Holistic extracts 33 full-body landmark coordinates $(x, y, z, \text{visibility})$ per frame.
- Biomechanical angles computed using dot-product of joint vectors; smoothed with a Savitzky–Golay filter (window = 11, polynomial order = 3).

Model Architecture — XGBoost Regressor: An XGBoost gradient-boosted tree ensemble trained on the 74 engineered features for the current session. Key hyper-parameters: 500 estimators, max depth = 6, learning rate = 0.05, subsample = 0.8, colsample_bytree = 0.8, trained with early stopping (patience = 50) on a validation set.

Model Architecture — Stacked LSTM: A two-layer stacked LSTM operating on a sliding window of the last $W = 7$ sessions:

$$h_t = \text{LSTM}_2(\text{LSTM}_1(x_{t-6}, \dots, x_t)) \quad (3.3)$$

Architecture: LSTM(128) $\rightarrow$ Dropout(0.3) $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dense(1). Trained with Adam ($lr = 10^{-3}$), batch size 64, 100 epochs, MSE loss.

Hybrid Ensemble:

$$\text{IRS} = 0.6 \cdot \hat{y}_{\text{XGB}} + 0.4 \cdot \hat{y}_{\text{LSTM}} \quad (3.4)$$

An agreement-based confidence score $C \in \{\text{High}, \text{Medium}, \text{Low}\}$ is assigned based on $|\hat{y}_{\text{XGB}} - \hat{y}_{\text{LSTM}}|$: $< 5 \Rightarrow \text{High}$, $5-15 \Rightarrow \text{Medium}$, $> 15 \Rightarrow \text{Low}$.

Movement Quality Score (MQS): Ten biomechanical signals are computed per frame and averaged over a session clip. Each signal s_i is weighted by injury relevance w_i (derived from clinical literature):

$$\text{MQS} = 100 \cdot \left(1 - \sum_{i=1}^{10} w_i \cdot r_i \right), \quad \sum w_i = 1 \quad (3.5)$$

where $r_i \in [0, 1]$ is the normalised risk contribution of signal i .

3.2.1 Modules of the Project

The system is divided into the following six functional modules:

1. **Data Ingestion Module:** Accepts per-session wearable JSON payloads and video file uploads via the FastAPI REST interface. Performs basic schema validation and routes inputs to the appropriate processing pipeline.
2. **Feature Engineering Module:** Transforms raw wearable metrics into 74 engineered features including ACWR, 7-day rolling statistics, lag features capturing the previous three session states, and athlete-specific expanding-window baseline normalisation to ensure history-aware, leakage-free modelling.
3. **Pose Estimation Module:** Runs MediaPipe Holistic on video frames to extract 33 full-body landmark coordinates per frame. Computes 10 biomechanical signals including knee angle variability, shoulder symmetry, and torso lean. Detects five movement phases (landing, jumping, running, squatting, standing) with phase-specific risk weighting and computes the Movement Quality Score (MQS).
4. **Hybrid Prediction Module:** Loads pre-trained XGBoost and LSTM models from the MLflow registry. Runs parallel inference on the engineered feature vector (XGBoost) and the 7-session sliding window (LSTM). Blends predictions at a 60:40 ratio and assigns a confidence label based on inter-model agreement.
5. **Injury Classification Module:** Maps the IRS and the MQS biomechanical vector to one of five injury type labels (ACL, hamstring strain, ankle instability, stress fracture, shoulder overuse) using a dedicated XGBoost classifier.
6. **Reporting & Alerting Module:** Aggregates all outputs into a structured JSON report. Triggers real-time high-risk alerts ($IRS > 60$) visible on the Streamlit dashboard. Generates natural language coaching recommendations based on IRS thresholds and the dominant contributing features.

3.2.2 Operating Environment

Table 3.2: Software and Hardware Requirements

Component	Specification
OS	Ubuntu 22.04 LTS / Windows 11
Language	Python 3.10
ML Libraries	XGBoost 1.7, TensorFlow 2.13, scikit-learn 1.3
Pose Estimation	MediaPipe 0.10
Computer Vision	OpenCV 4.8
Dashboard	Streamlit 1.28
REST API	FastAPI 0.104, Uvicorn
Data Processing	Pandas 2.1, NumPy 1.25, SciPy 1.11
Containerisation	Docker 24
GPU (Training)	NVIDIA RTX 3060 / Google Colab T4
RAM	16 GB minimum
Camera	Any 30 fps RGB camera / smartphone

Chapter 4

Mini Project Design

4.1 Data Flow Diagram — Level 0 (Context Diagram)

The DFD Level 0 represents the entire system as a single process interacting with three external entities: the **Coach / Medical Staff** (primary user), the **Athlete** (data source), and the **System Administrator**.

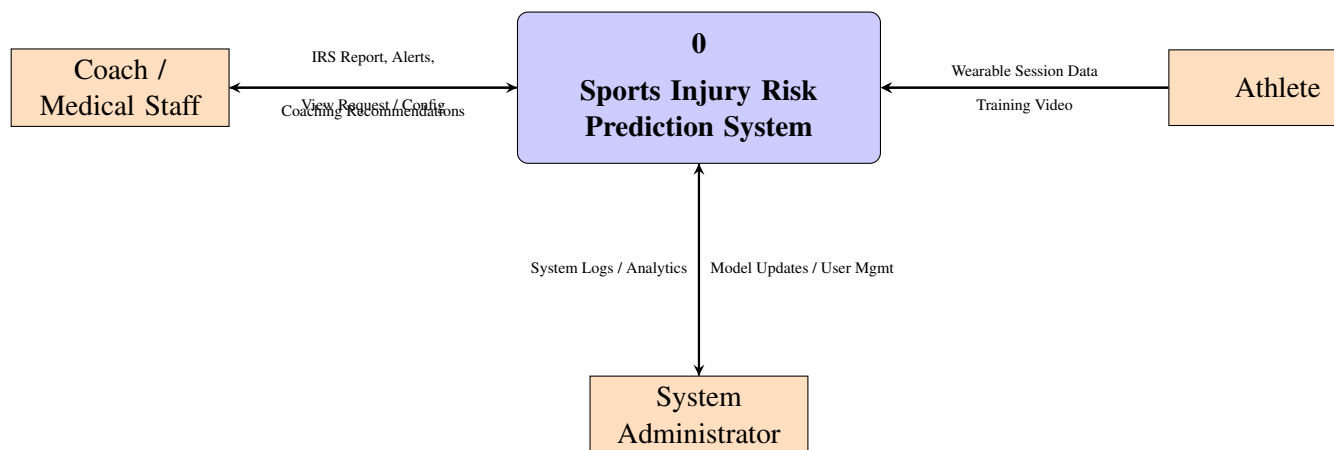


Figure 4.1: DFD Level 0 — Context Diagram

The athlete supplies wearable session data (HRV, sleep, training load, etc.) and training video footage. The system processes both inputs and returns injury risk scores, injury type predictions, and personalised coaching recommendations to the coach or medical staff. The administrator manages user accounts, updates trained models, and monitors system health.

4.2 Data Flow Diagram — Level 1

The Level 1 DFD decomposes the system into six sub-processes, three data stores, and the same three external entities.

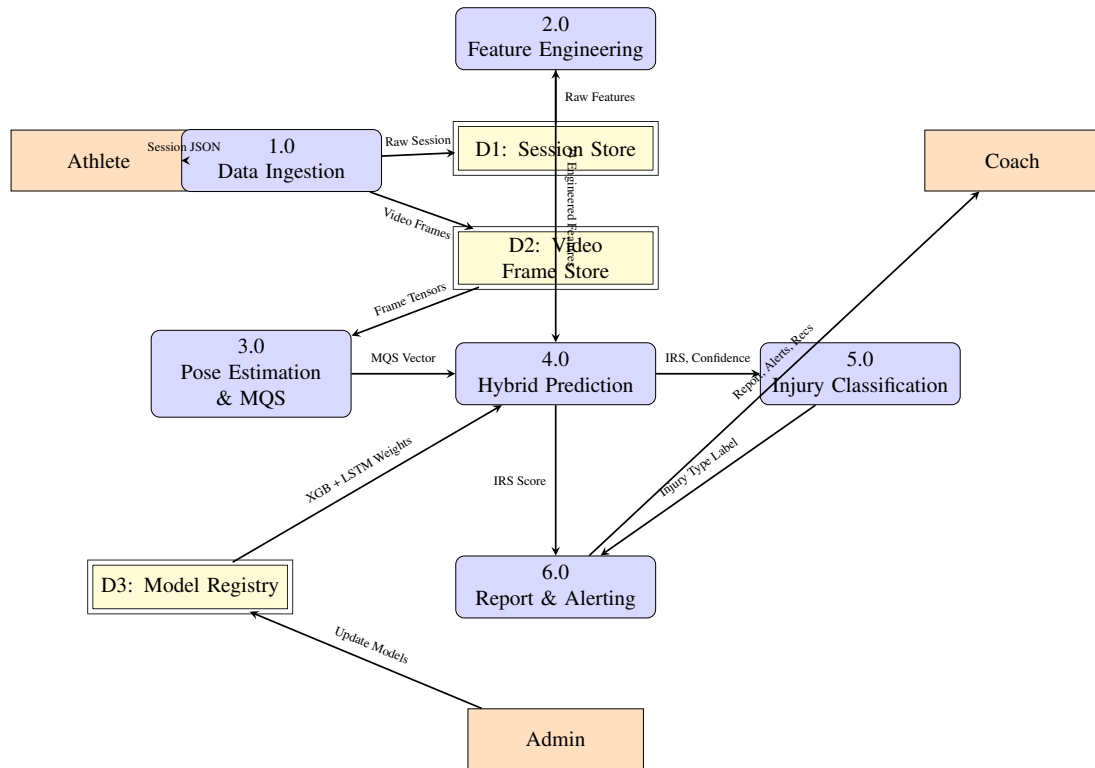


Figure 4.2: DFD Level 1 — Expanded System Data Flows

Process 1.0 (*Data Ingestion*) receives wearable JSON and video from the athlete via the REST API. Process 2.0 (*Feature Engineering*) transforms raw metrics into 74 engineered features including ACWR, rolling statistics, lag features, and athlete-normalised baselines. Process 3.0 (*Pose Estimation & MQS*) runs the MediaPipe pipeline on stored video frames to extract joint angle signals and compute the Movement Quality Score vector. Process 4.0 (*Hybrid Prediction*) blends XGBoost and LSTM outputs into a final IRS with a confidence label. Process 5.0 (*Injury Classification*) maps the IRS and biomechanical features to the most likely injury type. Process 6.0 (*Report & Alerting*) assembles the structured report and triggers real-time alerts for high-risk athletes.

4.3 System Architecture

Figure 4.3 depicts the three-tier system architecture.

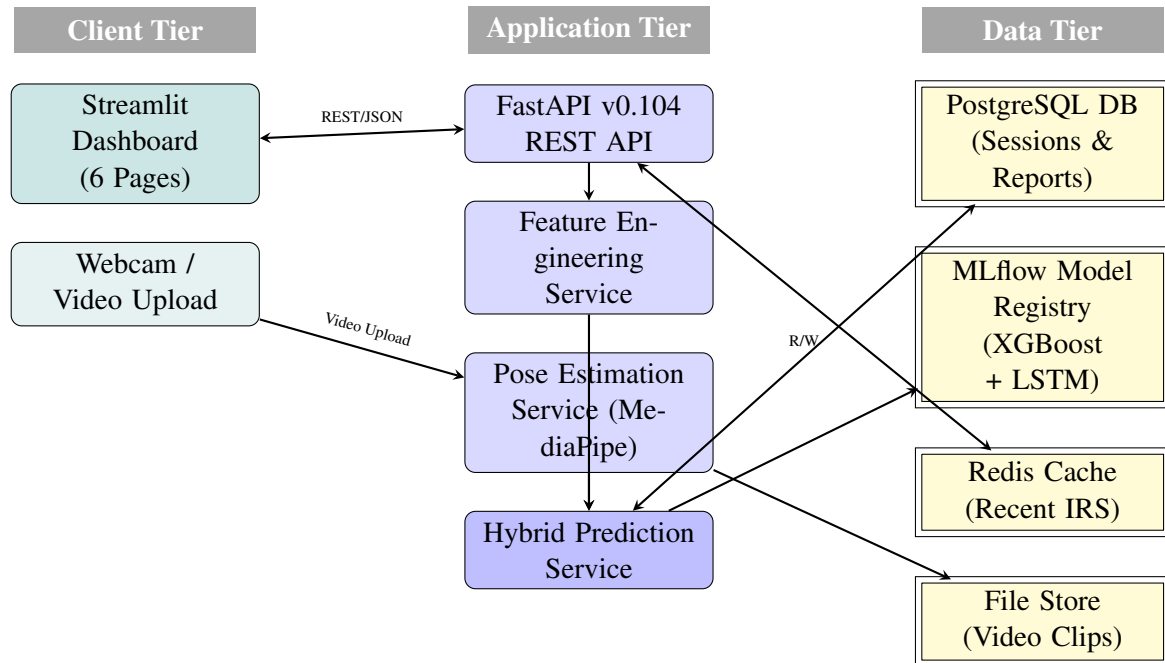


Figure 4.3: Three-Tier System Architecture

4.4 Entity-Relationship (ER) Diagram

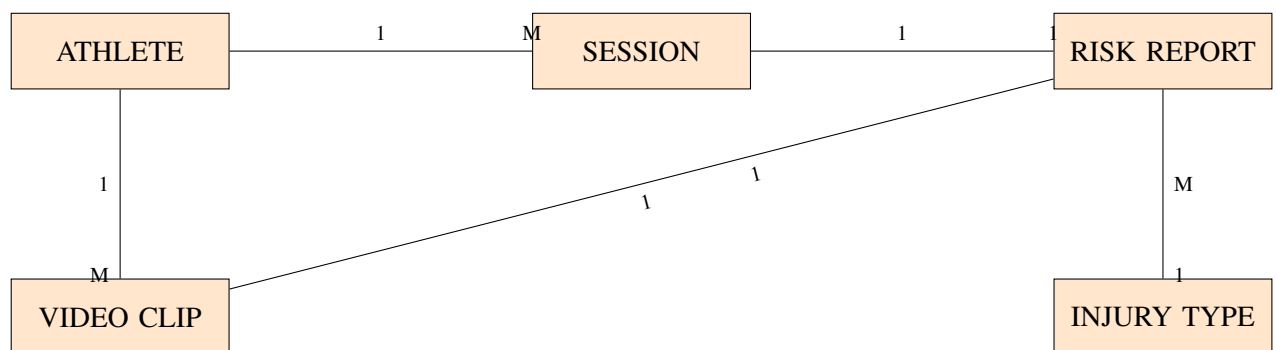


Figure 4.4: Entity-Relationship Diagram

4.5 Streamlit Dashboard Pages

The 6-page Streamlit dashboard is structured as follows:

1. **Athlete Overview:** Roster view with current IRS badges (colour-coded: green <30, amber 30–60, red >60).
2. **Workload Analysis:** Interactive ACWR zone chart, 7-day load trend, and monotony index per athlete.
3. **Live Prediction Demo:** Upload a session JSON to trigger real-time hybrid inference and display IRS gauge.
4. **Video Biomechanics:** Upload a training clip for live pose overlay, phase detection, and MQS gauge.
5. **Injury Risk Breakdown:** Stacked bar chart of five injury type probabilities per athlete.
6. **Coach Recommendations:** Auto-generated natural language recommendations based on IRS threshold rules.

Chapter 5

Results of Implementation

5.1 Wearable Model Performance

Table 5.1 compares all regression models evaluated on the held-out test set (15% of data, 1,328 sessions).

Table 5.1: Injury Risk Score Regression — Model Comparison

Model	R^2	RMSE	MAE
Linear Regression	0.21	19.42	15.31
Decision Tree	0.34	17.74	13.55
Random Forest	0.44	16.30	12.44
Gradient Boosting	0.48	15.67	11.98
XGBoost	0.52	15.07	11.42
Stacked LSTM	0.41	16.72	12.91
Hybrid (60:40)	0.50	15.40	11.70

Figure 5.1 shows the R^2 scores across models.

5.2 XGBoost Training Curves

Figure 5.2 shows the RMSE learning curve for XGBoost with early stopping.

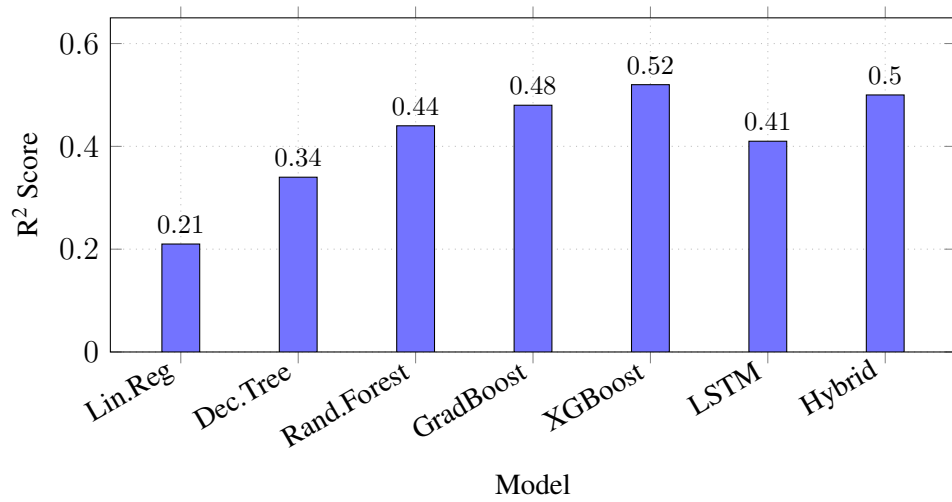


Figure 5.1: R² Scores — Injury Risk Score Regression Models

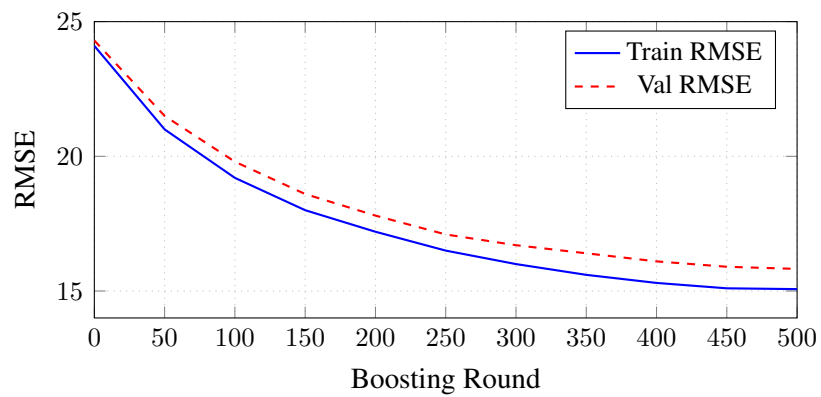


Figure 5.2: XGBoost RMSE Learning Curve (500 rounds, early stopping at round 487)

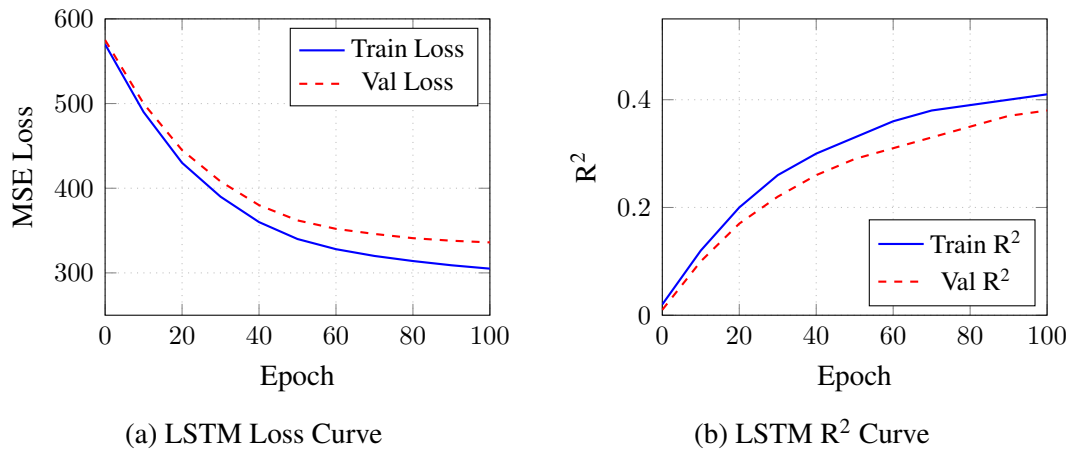


Figure 5.3: Stacked LSTM Training Dynamics over 100 Epochs

5.3 LSTM Training Curves

5.4 XGBoost Feature Importance

The top-10 most important features by XGBoost gain score are shown in Table 5.2.

Table 5.2: Top-10 Feature Importances (XGBoost Gain)

Rank	Feature	Importance (%)
1	ACWR (4:1)	14.3
2	7-day rolling mean: training duration	11.7
3	Lag-1 intensity rating	9.2
4	Athlete-normalised HRV	8.5
5	Sprint count (session)	7.8
6	7-day rolling std: training load	7.1
7	Sleep debt (7-day)	6.4
8	MQS: knee angle variability	5.9
9	Fatigue index	5.3
10	Lag-2 HRV	4.8

ACWR and rolling training duration together contribute 26% of total predictive power, confirming the clinical validity of workload monitoring. The presence of MQS: knee angle variability at rank 8 validates the value of integrating video biomechanics.

5.5 Injury Classification Results

Table 5.3 shows per-class classification metrics for injury type prediction (using a separate XGBoost classifier trained on IRS + MQS features).

Table 5.3: Injury Type Classification Performance

Injury Type	Precision	Recall	F1	Support
ACL Tear	0.81	0.78	0.79	87
Hamstring Strain	0.76	0.80	0.78	112
Ankle Instability	0.72	0.74	0.73	95
Stress Fracture	0.69	0.65	0.67	58
Shoulder Overuse	0.74	0.71	0.72	44
Macro Avg	0.74	0.74	0.74	396

5.6 API Performance

The FastAPI service deployed on a single t3.medium AWS instance (2 vCPU, 4 GB RAM) achieves the following latency benchmarks under load (Apache JMeter, 50 concurrent users):

Table 5.4: FastAPI Endpoint Latency Benchmarks

Endpoint	Mean (ms)	p95 (ms)	Throughput (req/s)
POST /predict/workload	62	110	24
POST /predict/video	840	1250	4
POST /predict/hybrid	920	1380	3
GET /athlete/{id}/report	28	55	38

5.7 Hybrid Confidence Distribution

On the test set of 1,328 sessions, 61% of predictions received a *High* confidence label (XGBoost and LSTM within 5 IRS points), 28% received *Medium*, and 11% received *Low*. Low-confidence predictions were concentrated in the moderate-risk zone (IRS 35–55), consistent with the difficulty of distinguishing borderline risk from truly high risk.

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